Basic Principles of Medicine 1

Module: Foundation to Medicine
Lecture: FTM 18

Enzymes: Inhibition and Regulation

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Objectives: Required reading: Properties of Enzymes

- Required reading before lecture ‘Enzymes’
- 🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402
  - Chapter 5: Enzymes: Sections: Properties; Mechanism of enzyme action; Factors affecting reaction velocity; Enzyme regulation
- Enzymes | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library

SOM.MK.I.BPM1.1.FTM.3.BCHM.0103	Describe the factors that influence enzyme reaction rates (substrate, pH and temperature).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0104	Discuss enzyme regulation via the concentrations of substrates or products.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0105	Describe enzyme regulation via the modulation of enzyme concentrations.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0106	Discuss conformation of enzymes and changes of conformation in relation to enzyme regulation. (reversible covalent modification, proteolytic cleavage and allosteric regulation).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0107	Describe the concept of proteolytic activation of digestive enzymes (eg: zymogens such as trypsinogen).

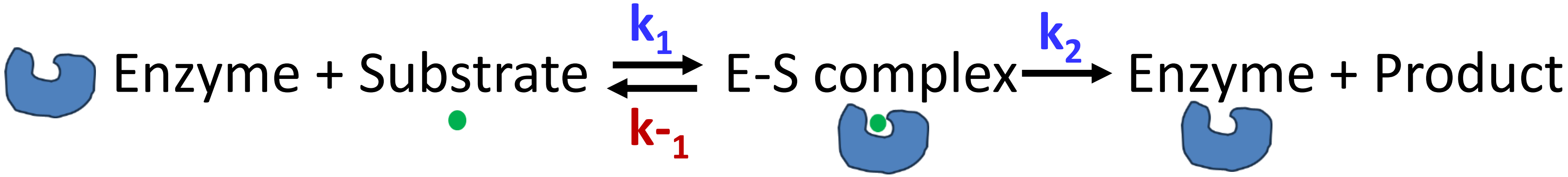
Objectives: Enzymes

SOM.MK.I.BPM1.1.FTM.3.BCHM.0108	Discuss the kinetics of allosteric enzymes and explain the sigmoidal graph.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0109	Describe allosteric enzymes and their regulation related to changes of $K_{0.5}$ or Vmax. Discuss in general the regulation of metabolic pathways related to feedback inhibition and feed-forward activation.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0110	Describe Michaelis-Menten kinetics
SOM.MK.I.BPM1.1.FTM.3.BCHM.0111	Define Km and Vmax and discuss their significance with the help of a graph.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0112	Discuss the inhibition caused by statin drugs (enzyme inhibited, effect on kinetics).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0113	Compare and contrast the mode of action and kinetics of reversible competitive and noncompetitive inhibitors.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0114	Compare and contrast the Michelis-Menten and Lineweaver-Burk plots.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0115	Indicate X- and Y- intercepts in the Lineweaver-Burk plot
SOM.MK.I.BPM1.1.FTM.3.BCHM.0116	Compare and contrast competitive and noncompetitive inhibition using Michaelis-Menten graph and Lineweaver-Burk plot.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0117	Distinguish between reversible and irreversible inhibitors.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0118	Discuss inhibition caused by DFP and aspirin (enzymes inhibited, effects of inhibition).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0119	Discuss suicide inhibition of enzymes using allopurinol as example.

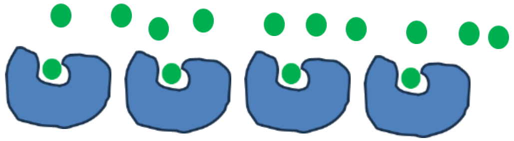
Recommended Reading

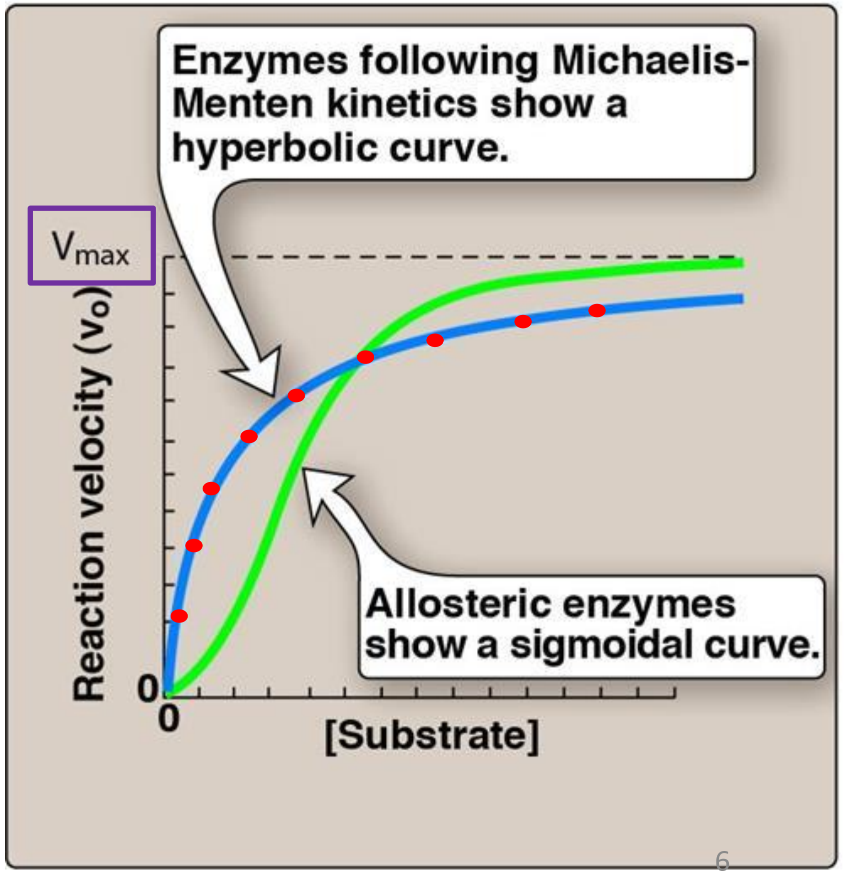
- 🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402; Chapter 5: Enzymes
- Lippincott Illustrated Reviews (**Lippincott**) 9th edition; Pages 57 – 68 and concept map page 73; Enzymes | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library
- Mark's Basic Medical Biochemistry: A clinical approach 6th edition;
  - Enzymes as Catalysts | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library
  - Regulation of Enzymes | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library
- Answers to case studies in Practice questions

Enzyme Kinetics: Michaelis Menten Kinetics (Vmax)

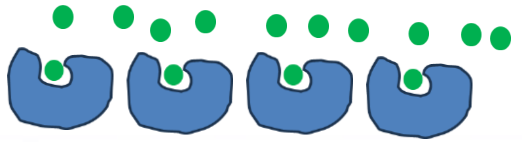


- Substrate binds to active site of an enzyme → Forms ES complex → Dissociates to form Product + Enzyme released
- When Substrate concentration [S] is plotted vs reaction velocity → Hyperbolic curve (Michaelis-Menten kinetics)
- **Vmax (Maximal velocity)**: When **ALL** active sites on enzyme saturated (filled) with substrate

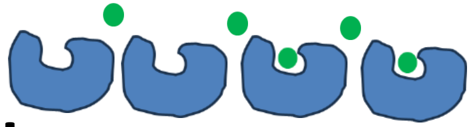




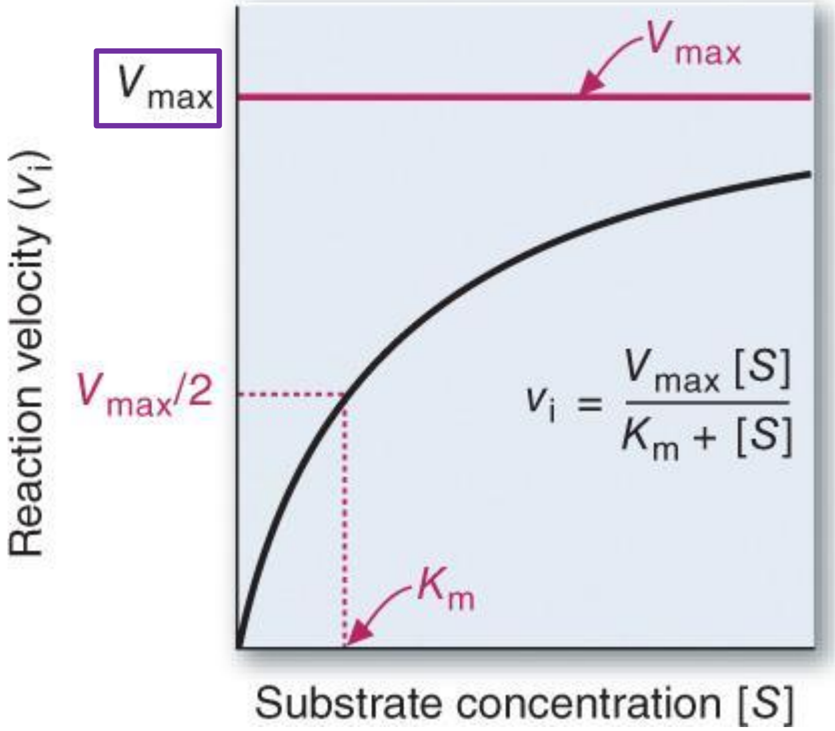
Enzyme Kinetics: Michaelis Menten Kinetics (Km)



- **Km (Michaelis constant): Substrate concentration at ½ maximal velocity (½ Vmax)**
- [S] at which half **(50%)** of enzyme's active sites are filled with substrate

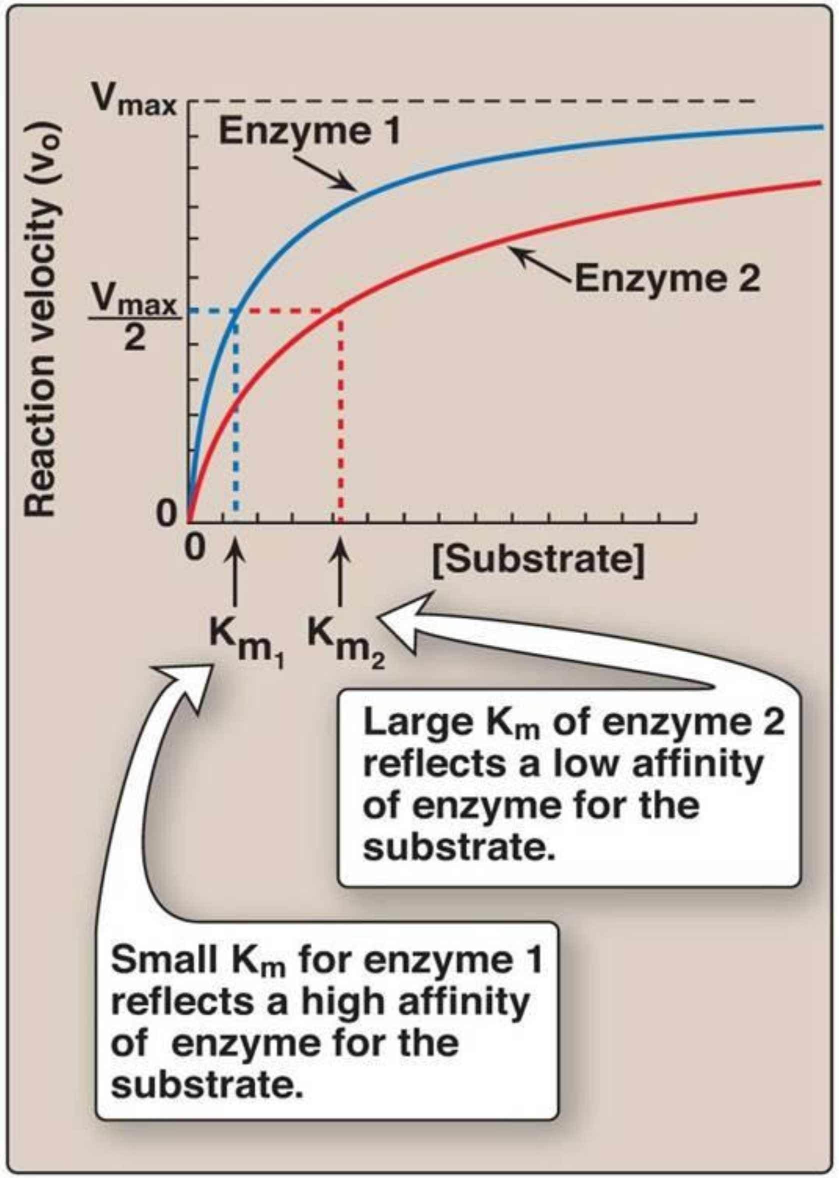


- **Reflects affinity of enzyme for substrate**
  - **Km inversely related to affinity for substrate**
- Km = **(k_{-1} + k_2)/k_1**
- Michaelis Menten equation links velocity (v_o) and [S]
  - $v_0 = \frac{V\max\ [S]}{Km+[S]}$

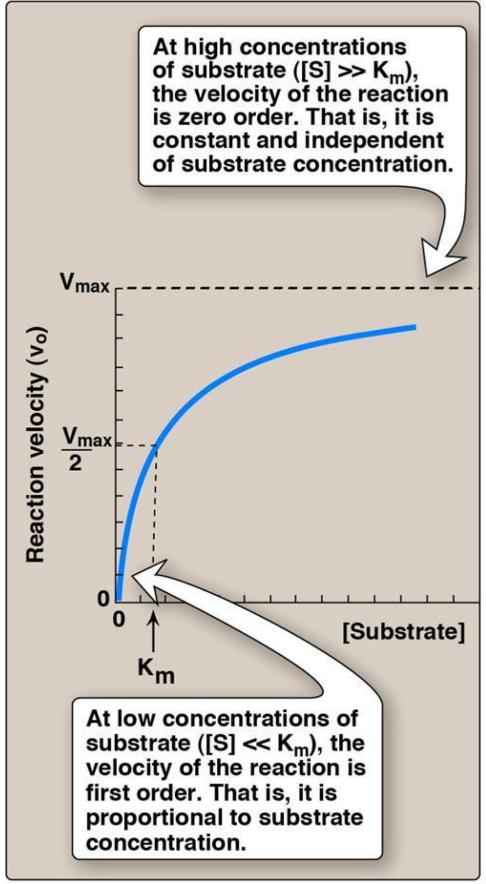


Km and Substrate affinity

- **Km inversely related to substrate affinity**
- Higher Km $\Rightarrow$ Lower substrate affinity
- Lower Km $\Rightarrow$ Higher substrate affinity
- Compare substrate affinities of enzymes 1 and 2 in the graph
- **Inhibitors affect kinetics (Km and Vmax)**
- **In many inherited enzyme deficiency disorders, the defective (inefficient) enzyme may have higher Km (need higher [S] to reach ½ Vmax) or lower Vmax**

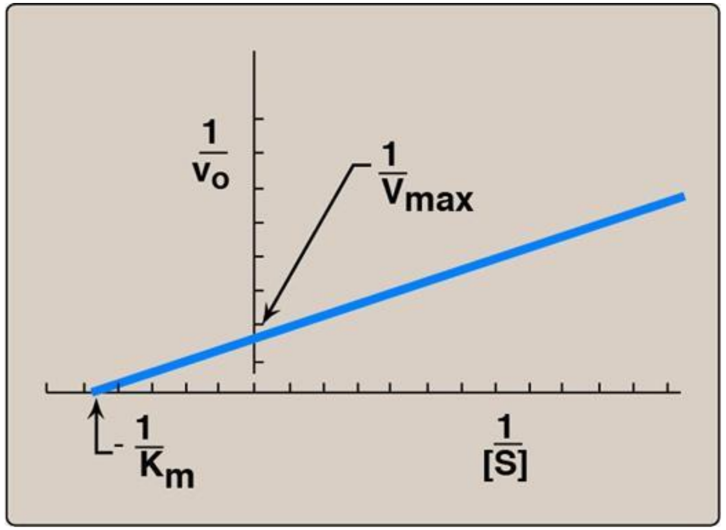


Michaelis-Menten and Lineweaver-Burk plots



Michaelis-Menten graph

- **Hyperbolic curve**
- **Vmax and Km estimated from graph (Not very accurate)**



Lineweaver-Burk plot

- **Straight line**
- **Vmax and Km mathematically obtained (More accurate)**
- **X-intercept: -1/Km**
- **Y-intercept: 1/Vmax**

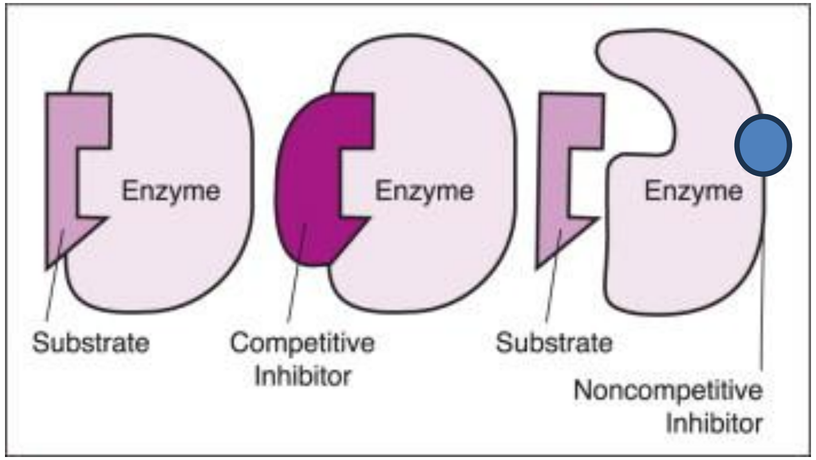
SOM.MK.I.BPM1.1.FTM.3.BCHM.0114; 0115

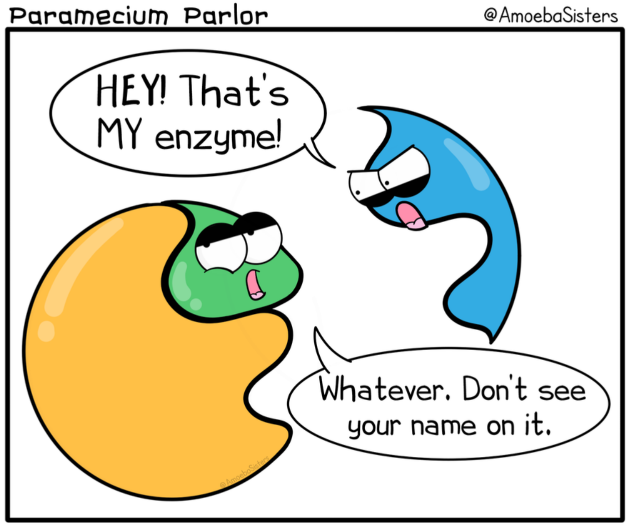
Enzyme inhibition

- Inhibitor binds to the enzyme and reduces product formation
- Toxins or therapeutic drugs
- Types of inhibition
  - Reversible **(weak, non-covalent links between enzyme and inhibitor)**
    - Competitive
    - Non-competitive
  - Irreversible **(Covalent links between enzyme and inhibitor)**
  - Suicide inhibition

Competitive inhibition (Reversible)

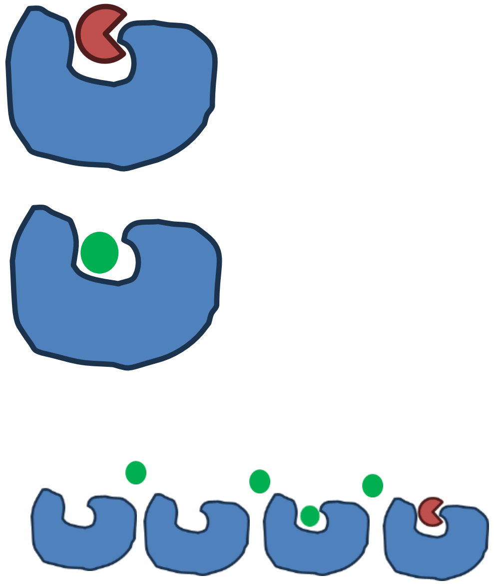
- Inhibitor (I) structural analog of substrate 
- Binds to enzyme's **active site (competes with S)**
  - Non-covalent interactions
- Reduces ES complex formation
- EI complex → No product formed
- **Inhibition reversed when [Substrate] is very high → Vmax unchanged**
- **Apparent Km increased**
  - **More [S] required to reach Vmax/2**
  - **Apparent affinity for substrate decreases**

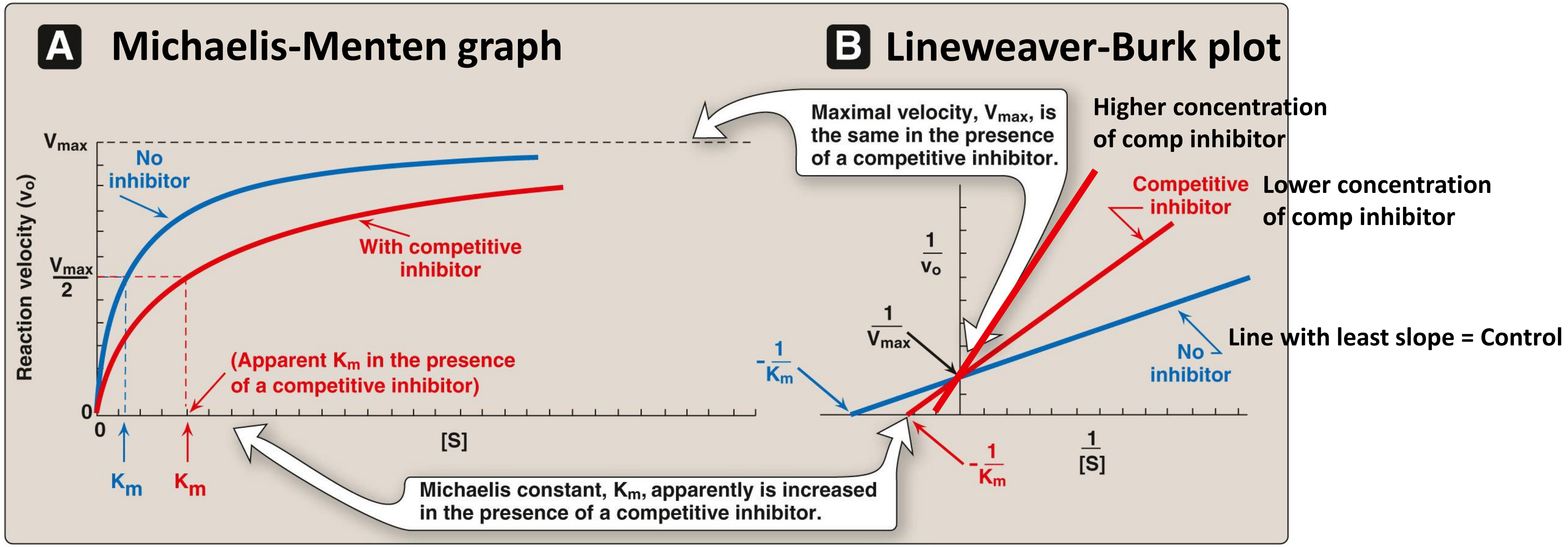




Competitive Inhibitors: If it fits, it sits.

Competitive inhibition (Reversible): Kinetics





- **Inhibition reversed when [Substrate] is very high → Vmax unchanged**
- **Apparent Km increased**
  - **More [S] required to reach Vmax/2**
  - **Apparent affinity for substrate decreases**
- **Lines intersect at 1/Vmax (Lineweaver-Burk plot)**

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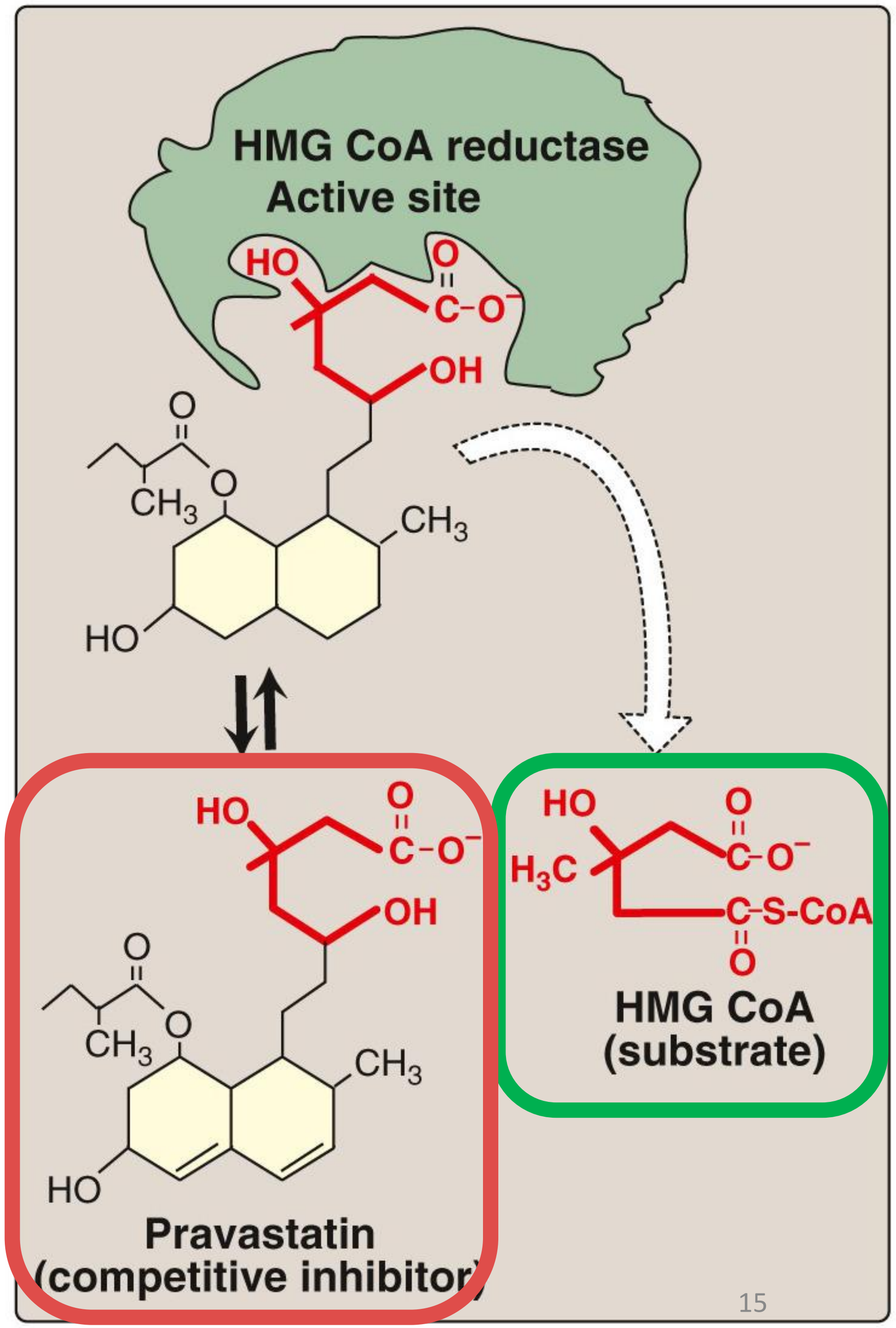
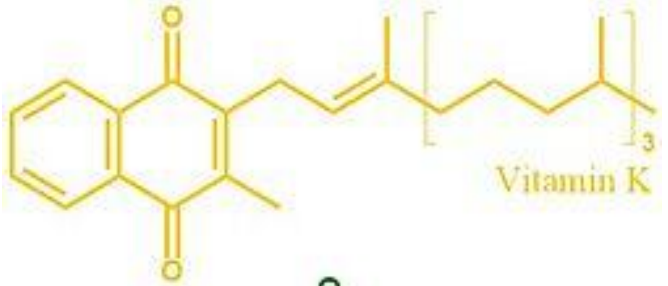
Competitive inhibition (Reversible): Example

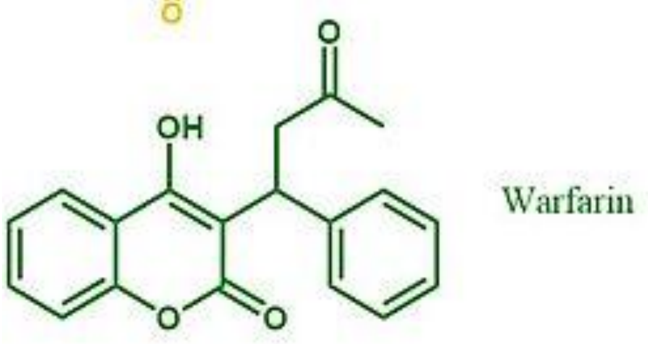
- HMG CoA reductase (regulated enzyme of cholesterol biosynthesis pathway)
- Statins used to reduce cholesterol synthesis and serum cholesterol levels
  - HMG-CoA analog (substrate)
  - Bind to active site
  - Increases apparent Km and Vmax unchanged



Competitive inhibition (Reversible): High [S]

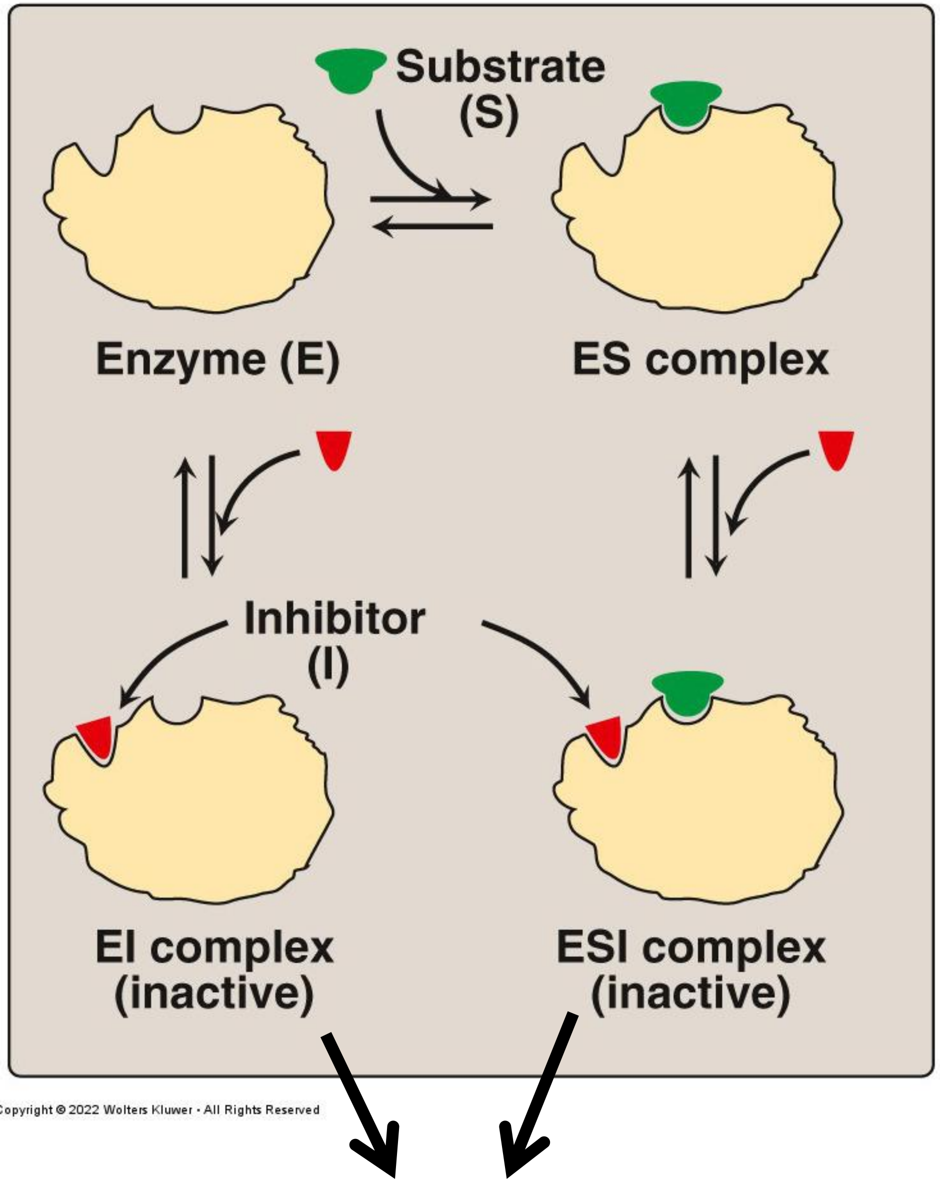
- Warfarin (anticoagulant) **competitively** inhibits Vitamin-K mediated modification of clotting factors
- Warfarin: Substrate analog of Vitamin K (substrate)
- Warfarin toxicity: Treat with High doses of vitamin K (substrate) → Reverses inhibition
- Do NOT memorize structure





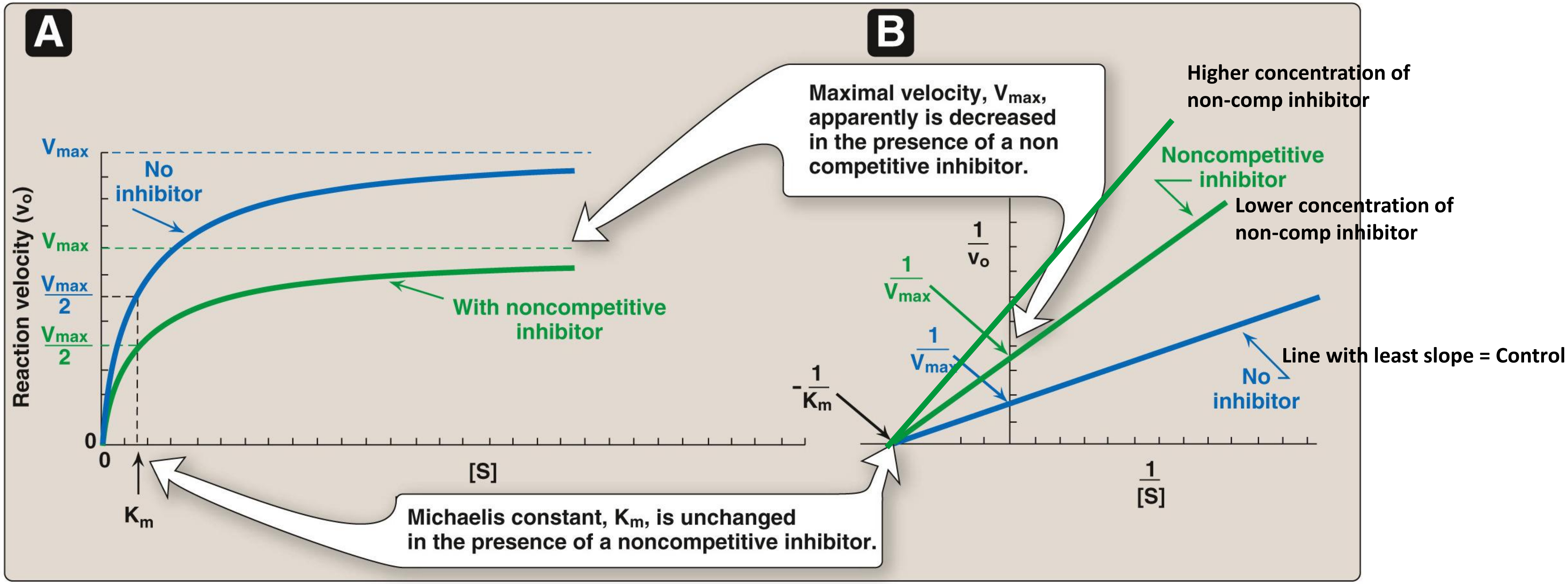
Non-competitive inhibition (Reversible)

- Inhibitor binds to a site other than active site
- Binds to E or ES complex and inhibits product formation
  - Non-covalent interactions
- **Apparent Vmax is lower**
- **Km unchanged**
- Increasing substrate concentration does NOT reverse inhibition

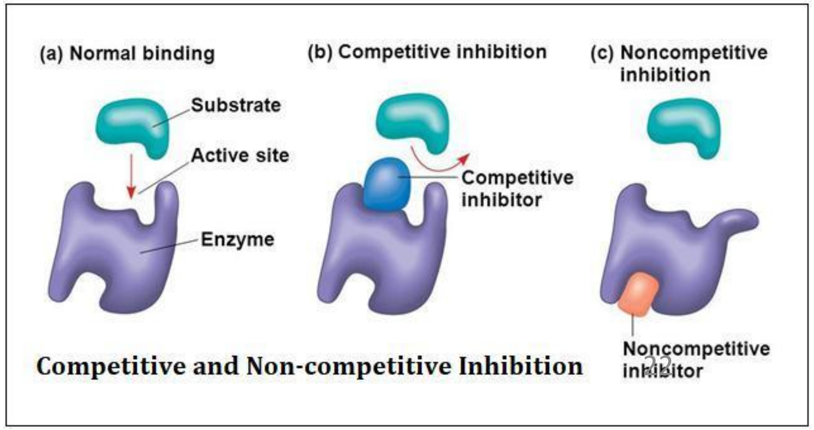


NO PRODUCT FORMED

Non-competitive inhibition (Reversible): Kinetics



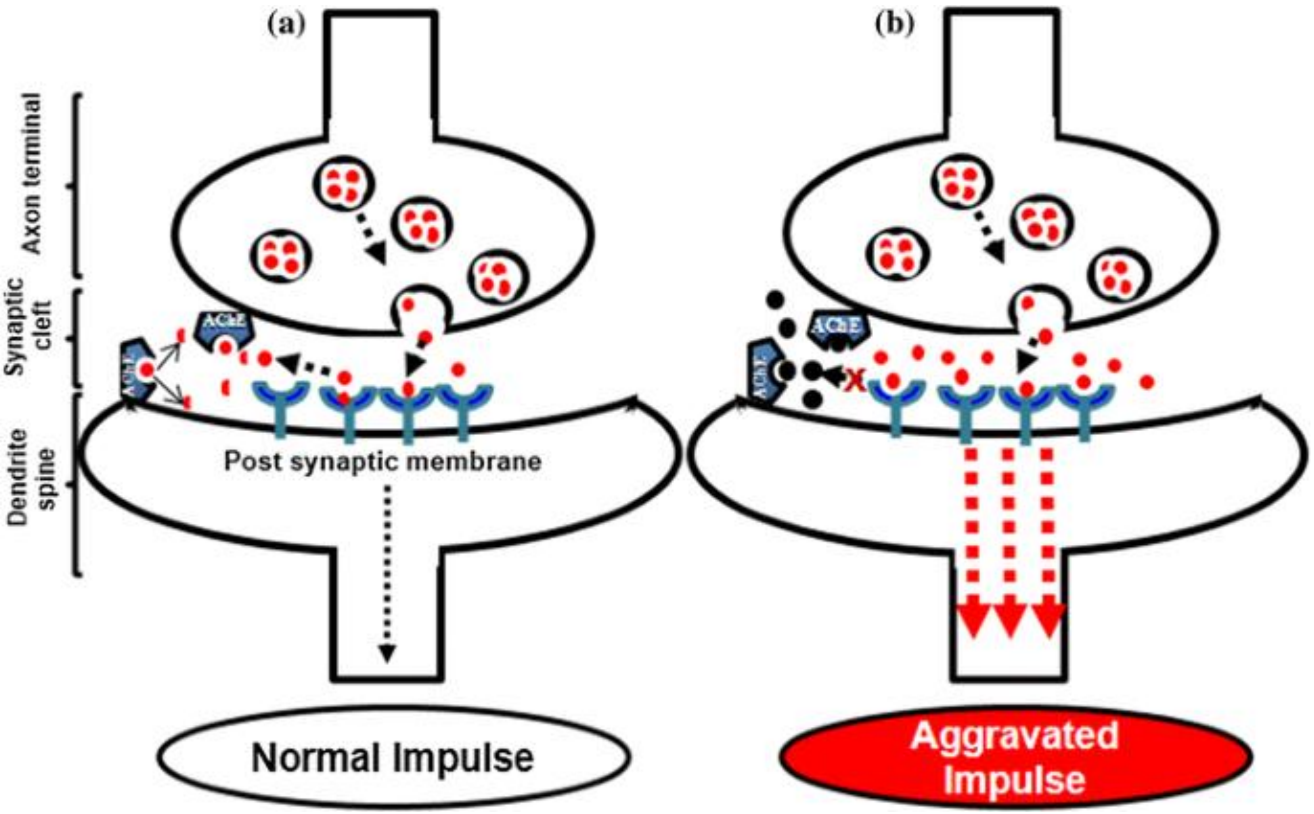
- Apparent Vmax is lower
- Km unchanged
- **Lines intersect at -1/Km (Lineweaver-Burk plot)**

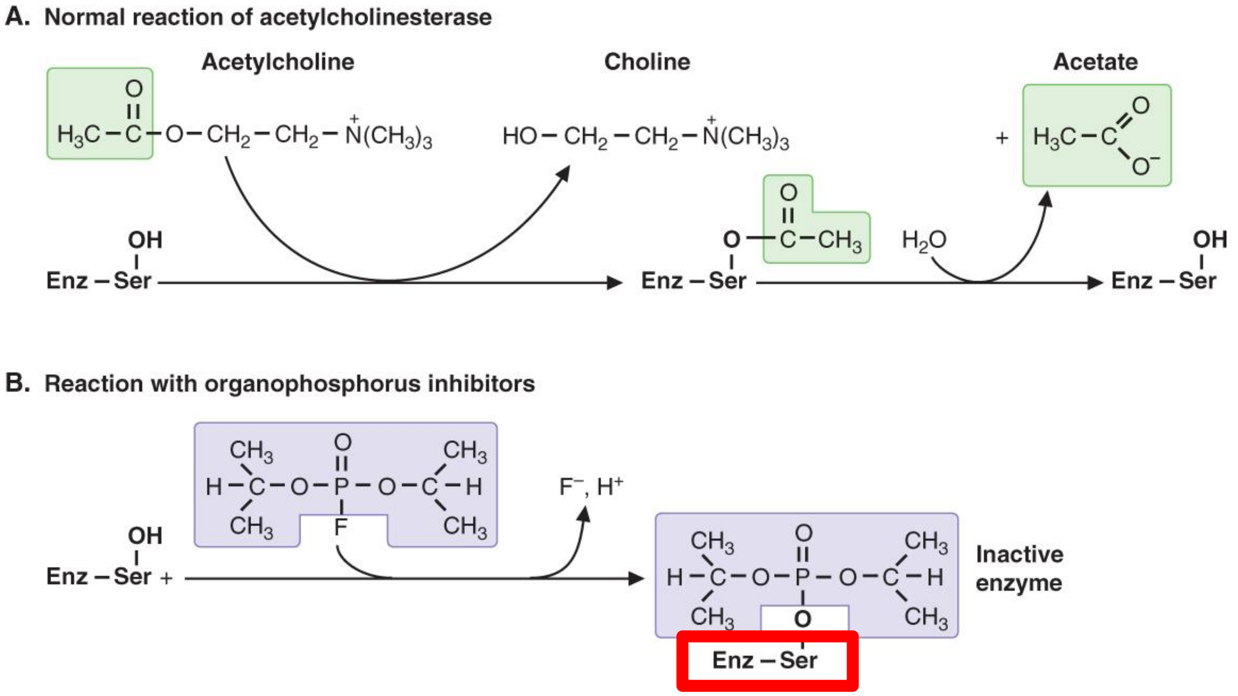


Irreversible inhibition

- Inhibitor binds to enzyme by **covalent links**
- Enzyme-Inhibitor complex → Degraded
- Inhibition overcome by **“synthesis of new enzyme”**
- Examples:
  - Acetylcholinesterase inhibition by DIFP (organophosphate/insecticides)
  - Cyclooxygenase inhibition by aspirin

Irreversible inhibition: DIFP (organophosphorus pesticides)

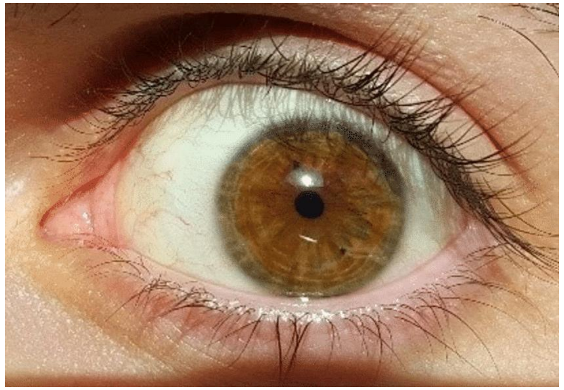




- **Acetylcholinesterase** cleaves acetylcholine (neurotransmitter) → Choline + Acetate
- Organophosphate insecticides irreversibly inhibit acetylcholinesterase → Higher acetyl choline levels → Aggravated cholinergic impulses

Irreversible inhibition: DIFP (organophosphorus pesticides)

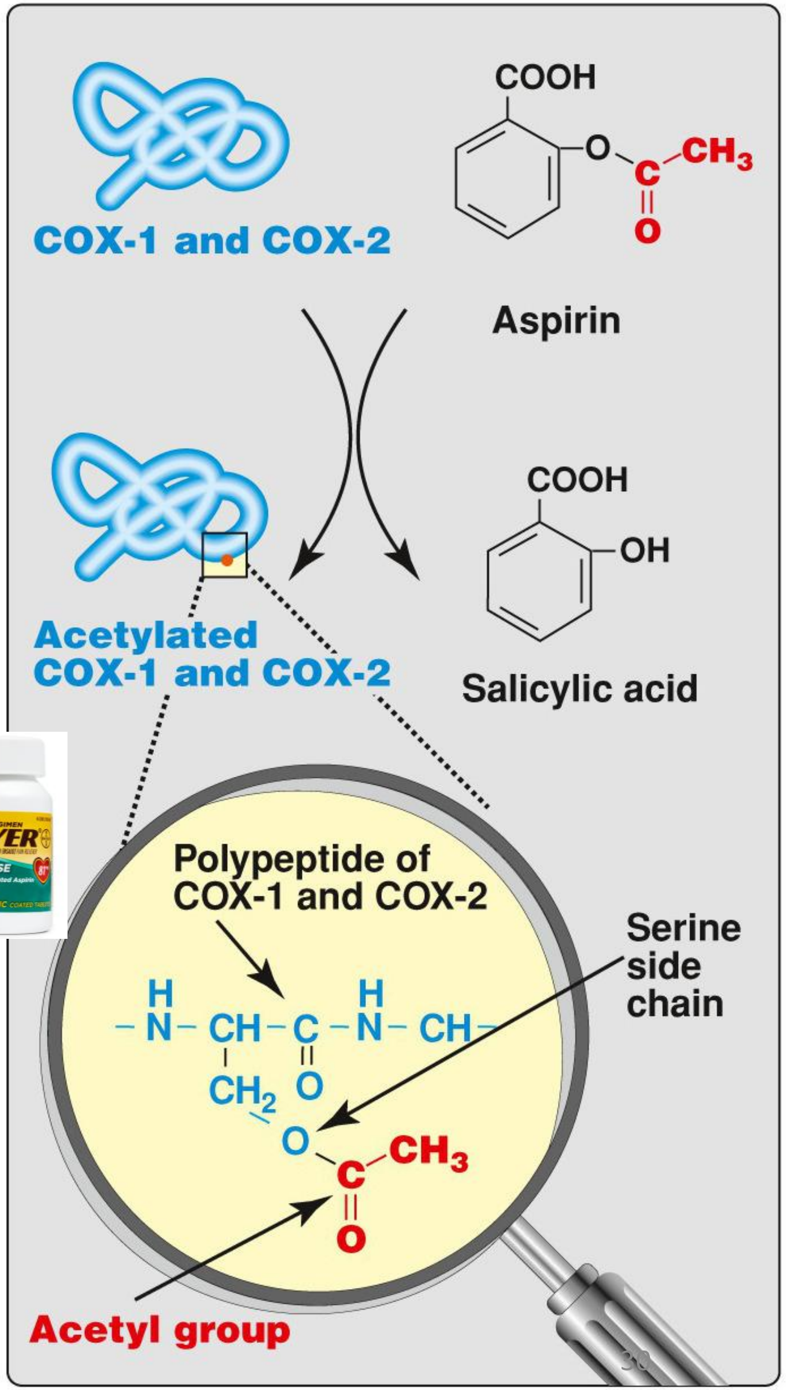
- Organophosphate insecticides (DFP) **irreversibly** inhibit **acetylcholinesterase**
  - Nerve gas (sarin), malathion have similar effects
- DIFP forms **covalent bond** with **Serine residue** in active site
- Higher acetylcholine levels in synapses
  - Constriction of pupils
  - Sweating
  - Bronchoconstriction and difficulty breathing



Pinpoint pupils

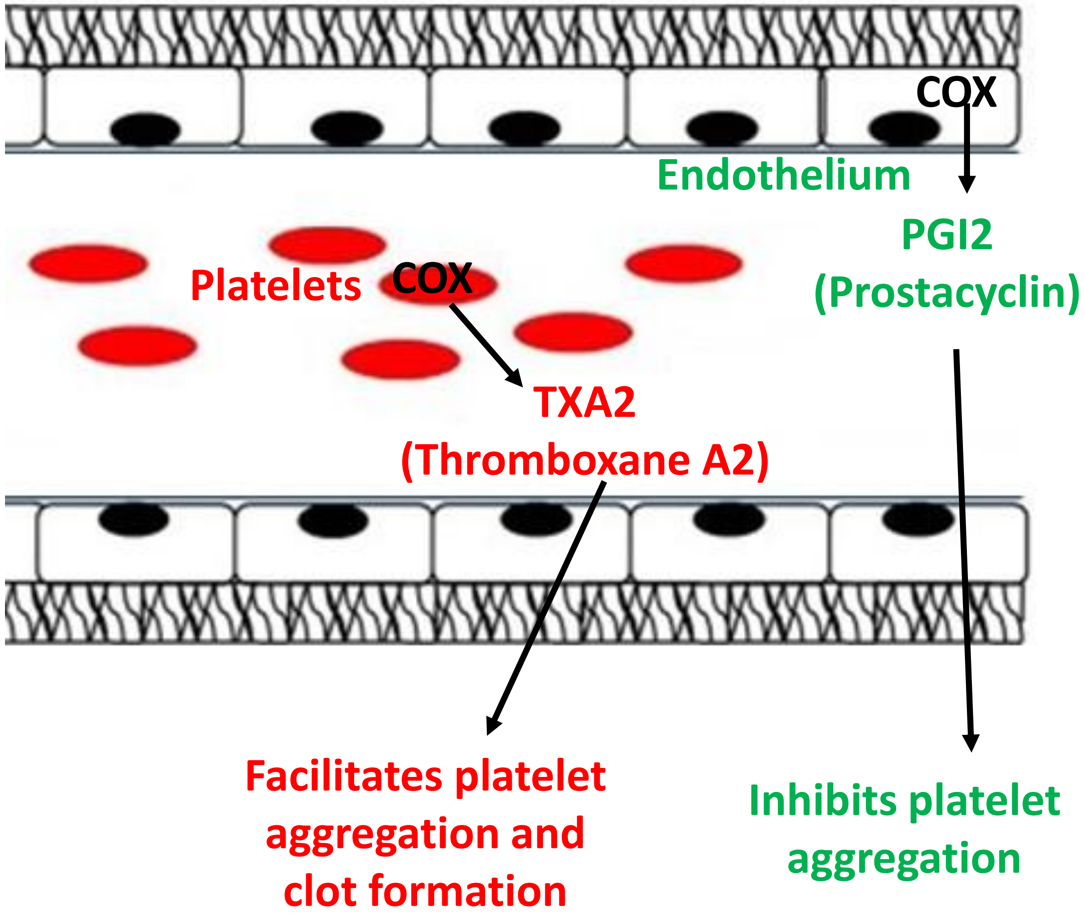
Irreversible inhibition: Aspirin

- Aspirin = Acetyl salicylic acid (Non-steroidal anti-inflammatory drug; NSAID)
- Inhibits cyclooxygenase (COX-1 and COX-2)
- Adds **'acetyl'** group to **Serine** at active site
- **Irreversible** inhibition
- Low-dose aspirin (81 mg): Anti-thrombogenic (anti-platelet drug)
- Used as an anti-inflammatory drug (higher dose)



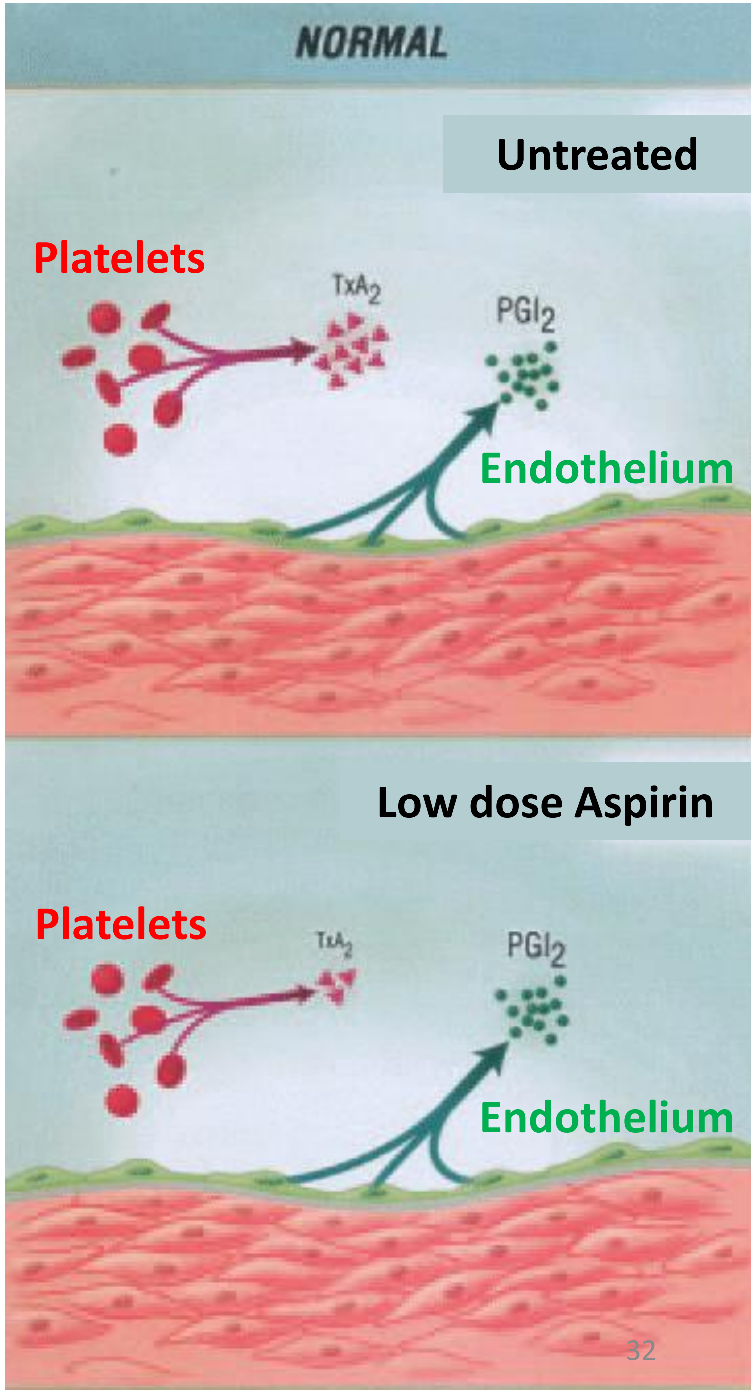
Role of COX: Platelets and endothelial cells

- Cyclooxygenase (COX) enzyme: Platelets and endothelial cells
- In platelets, COX forms **Thromboxane A2** (TXA2) from arachidonic acid → facilitates platelet aggregation and clot formation
- In endothelial cells, COX forms **Prostacyclin (PGI2)** → inhibits platelet aggregation
- **Higher TXA2: Prostacyclin (PGI2) ratio ⇒ higher tendency to form clots**
- **Higher PGI2: TXA2 ratio ⇒ lower tendency to form clots**



Irreversible inhibition: Aspirin

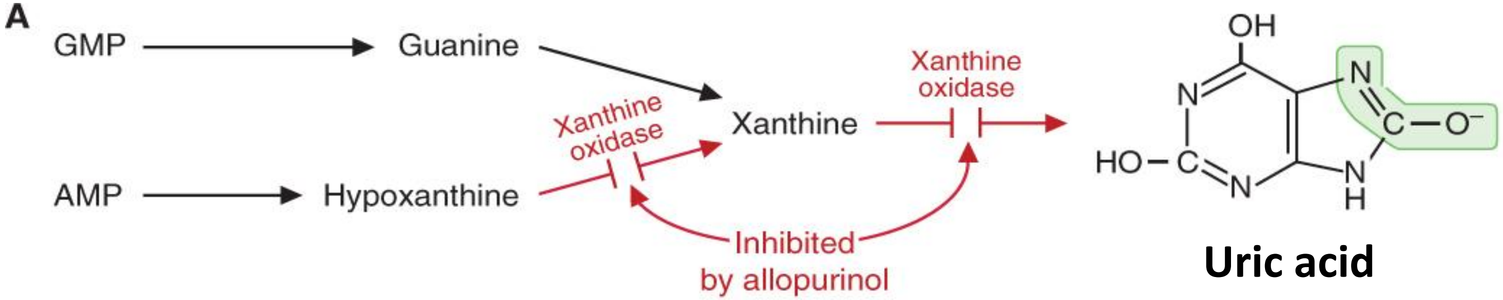
- Aspirin inhibits COX → decreases TXA2 formation in **platelets**
  - Platelets **do NOT have nucleus;** CANNOT synthesize new enzyme
- PGI2 synthesis not affected
  - Endothelial cells have nucleus and synthesize new COX enzyme
- **Higher PGI2: TXA2 ratio ⇒ lower tendency to form clots**
- Low-dose aspirin: **Anti-thrombogenic** (Anti-platelet drug)



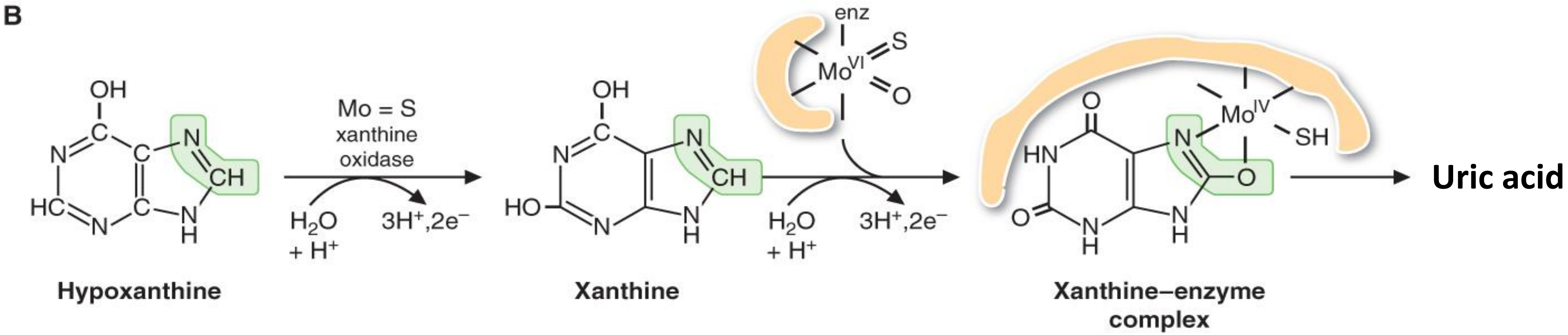
Suicide inhibition: Allopurinol

- Inhibitor binds to **active site** → Undergoes chemical modifications → Forms **irreversible links** to enzyme (Irreversible inhibitor)
- Example: Allopurinol inhibits **Xanthine oxidase** **(Purine nucleotide degradation)**
  - **Xanthine oxidase converts**
    - **Hypoxanthine to xanthine and**
    - **Xanthine to uric acid**
  - Allopurinol binds to active site of xanthine oxidase → Undergoes part of reaction (modified by enzyme to **alloxanthine/oxypurinol**) → Binds irreversibly to enzyme's active site (Irreversible inhibition)
  - Allopurinol reduces Uric acid formation
  - Used in **Gout** (Elevated uric acid levels→ Joint pain)

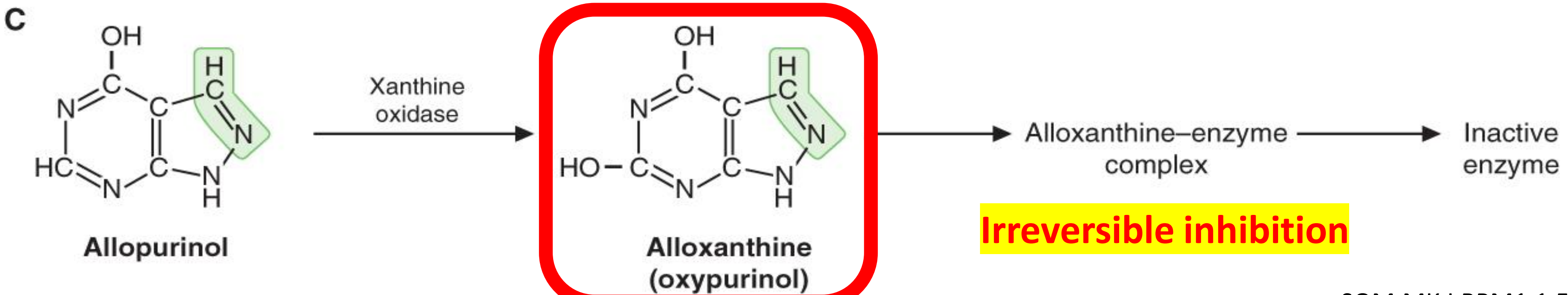
Suicide inhibition (mechanism-based): Allopurinol



Purine nucleotide degradation



Normal mechanism of action of Xanthine oxidase



Irreversible inhibition

Suicide inhibition of Xanthine oxidase by Allopurinol

Regulation of Enzymes

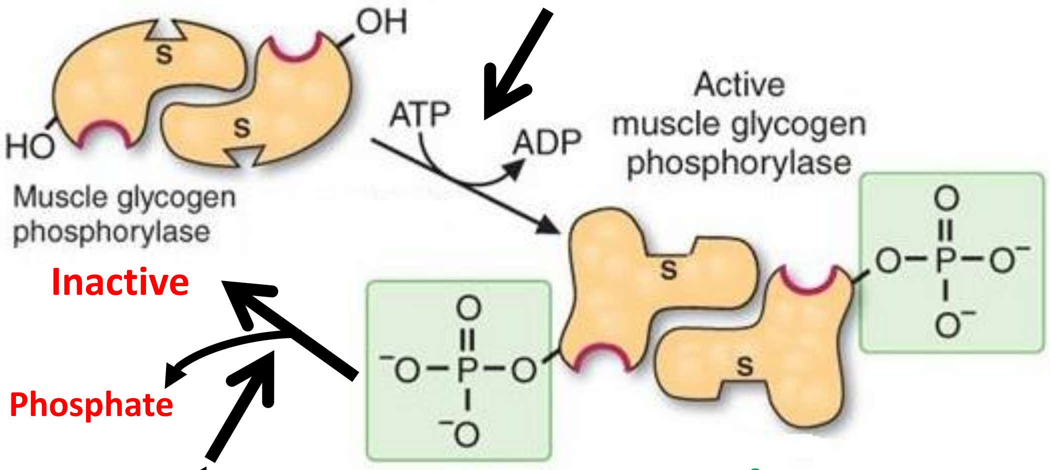
- Flux through biochemical pathways can be altered under different physiological states to respond to different needs of cells and organisms
- Different types of regulation: **Differentiate and explain**
  - Short-term (Response in minutes)
    - Covalent modification of enzymes
    - Allosteric modulation of enzymes
    - Changing concentration of substrates or products
    - Irreversible proteolytic activation
  - Long-term (Response in hours to days)
    - Induction and repression (Change enzyme concentration by altering rates of synthesis)

Short-term regulation of Enzymes: **Covalent modification**

- Enzyme activity altered by adding/ removing phosphate groups
- **Addition of Phosphate** to Serine or Threonine side chains → Increases/ decreases activity of regulatory enzymes
- Acts like “SWITCH” → Increase/ decrease activity
- Glucagon/ Epinephrine → Phosphorylation → **Enzyme conformation change** → Rapid change in enzyme activity → Increases glycogen breakdown
- Insulin → Removal of phosphate → Enzyme less active → Decreases glycogen breakdown
- Reversible (respond to different signals)
- Refer to signal transduction lecture

Glucagon/ Epinephrine

Protein kinase

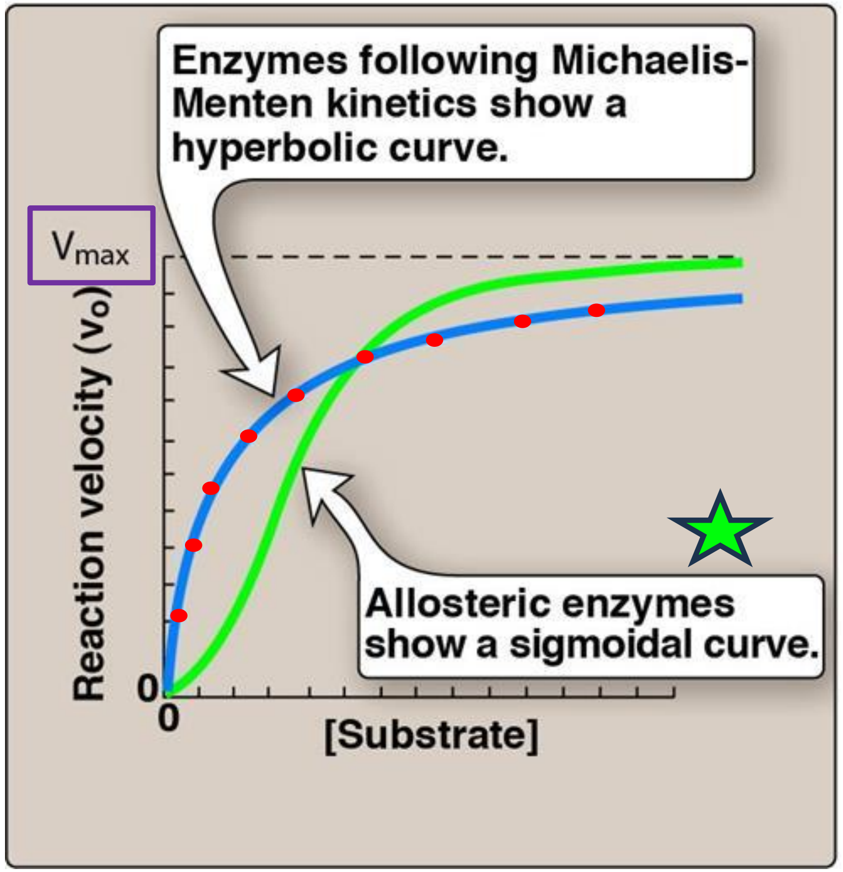


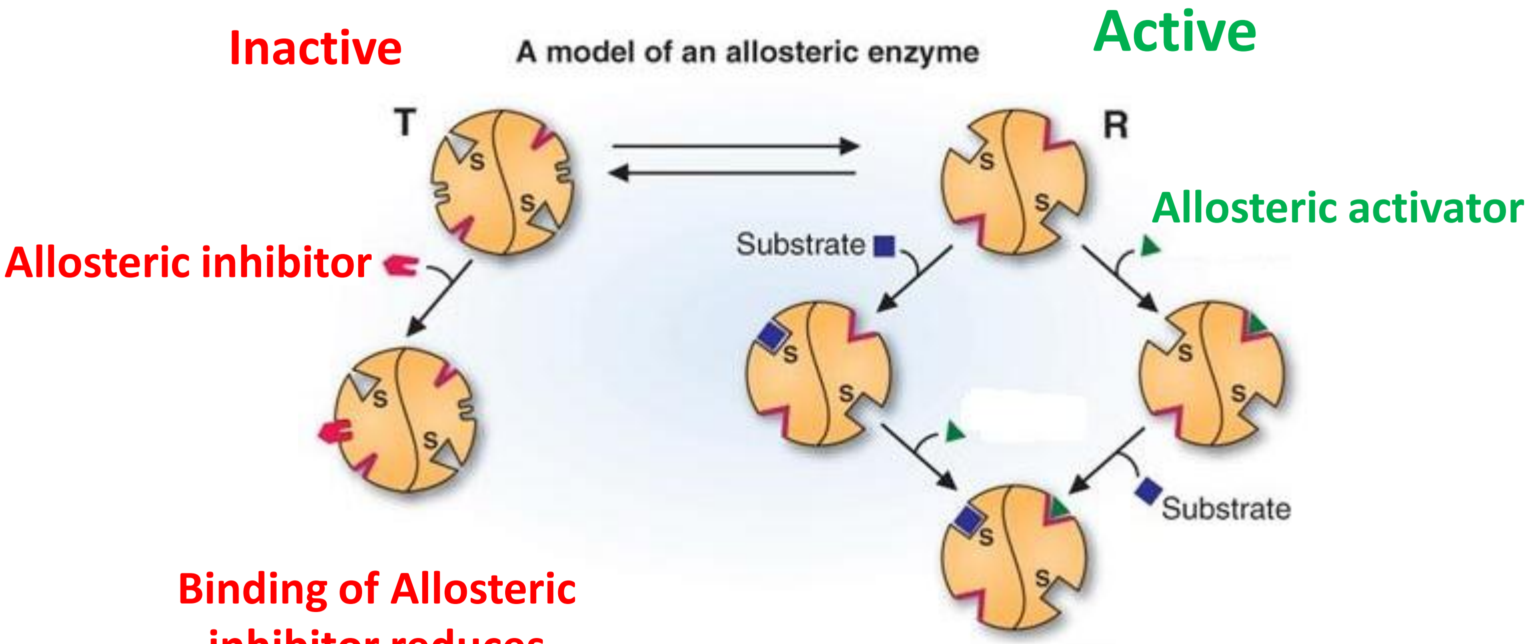
Insulin

Active

Increases glycogen breakdown

Short-term regulation of Enzymes: **Allosteric modulation**





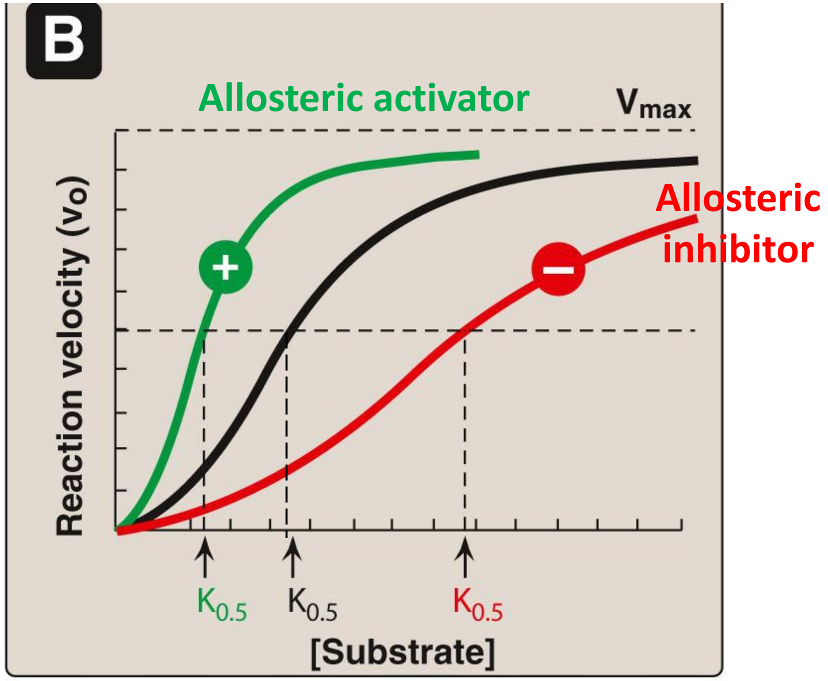
Binding of Allosteric inhibitor reduces substrate binding

Binding of Allosteric activator increases substrate binding → Increases Product formation

- Multi-subunit enzymes
- Modulator binds to site other than active site (**Allosteric site**) → **Conformational change in enzyme → Alters substrate affinity**

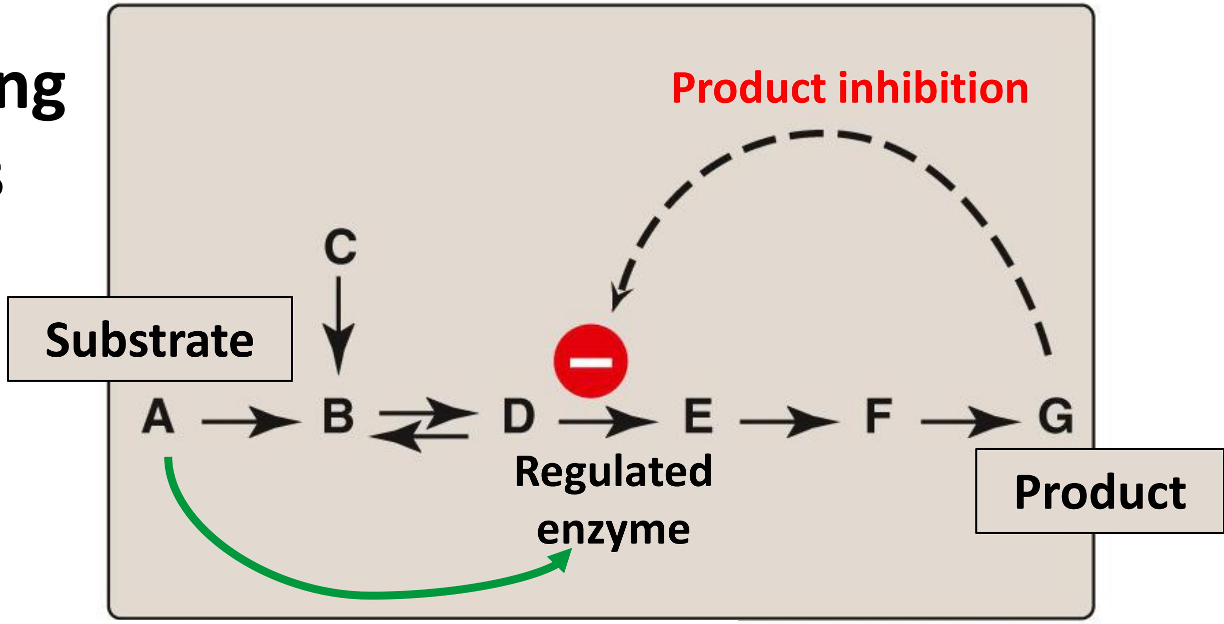
Short-term regulation of Enzymes: Allosteric modulation: **Kinetics**

- Allosteric enzymes: **Cooperative** substrate binding
- Substrate binding to one subunit increases substrate binding to other subunits → **Sigmoid saturation kinetics** (S-shaped)
- Michaelis-Menten kinetics: Hyperbolic curve
- $K_{0.5}$: Substrate concentration where velocity is Vmax/2 **(Compare to Km)**
- **Allosteric inhibitor** → Increases $K_{0.5}$ → **Decreases substrate affinity (Right shift)**
- **Allosteric activator** → Decreases $K_{0.5}$ → **Increases substrate affinity (Left shift)**
- Differentiate allosteric inhibitor (physiological regulator) vs Competitive and non-competitive inhibitors (pharmacological drugs/toxins)

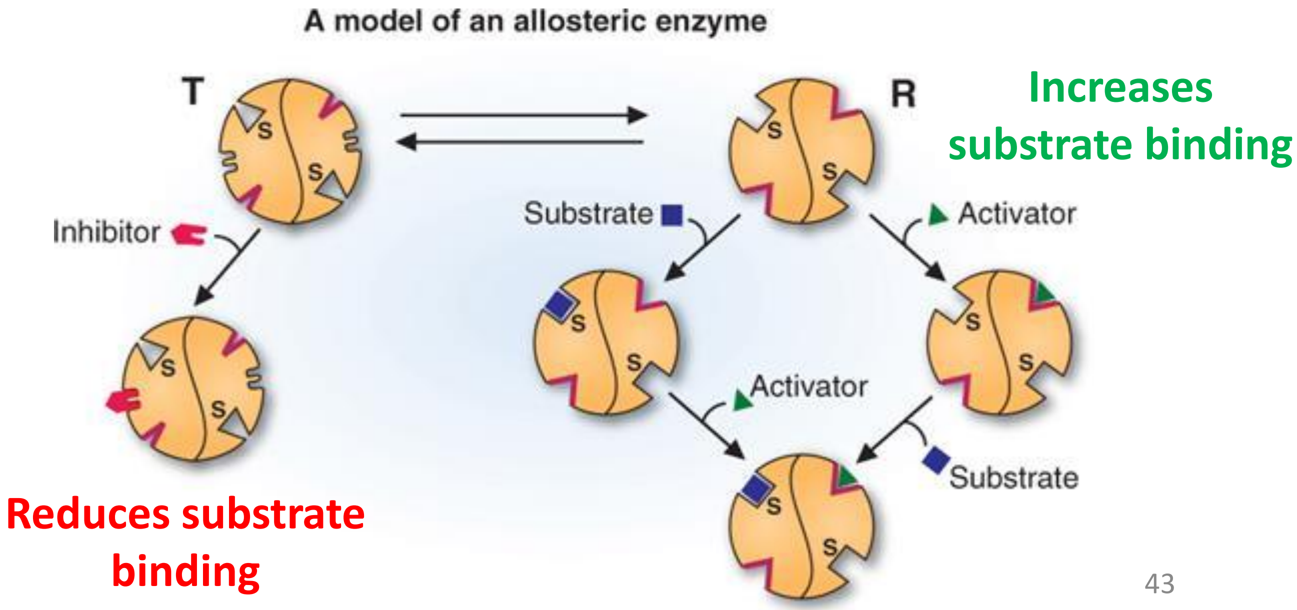


Short-term regulation of Enzymes: **Changing concentration of substrates or products**

- Regulated enzymes (Allosteric enzymes)
- Higher levels of **Product 'G'** → Bind to allosteric site of regulated enzyme → **Allosteric inhibitor** → Reduces pathway flux (**product inhibition**)
- Higher levels of **Substrate 'A'** → Binds to allosteric site of regulated enzyme → **Allosteric activator** → Increased flux through pathway **(feed-forward activation)**

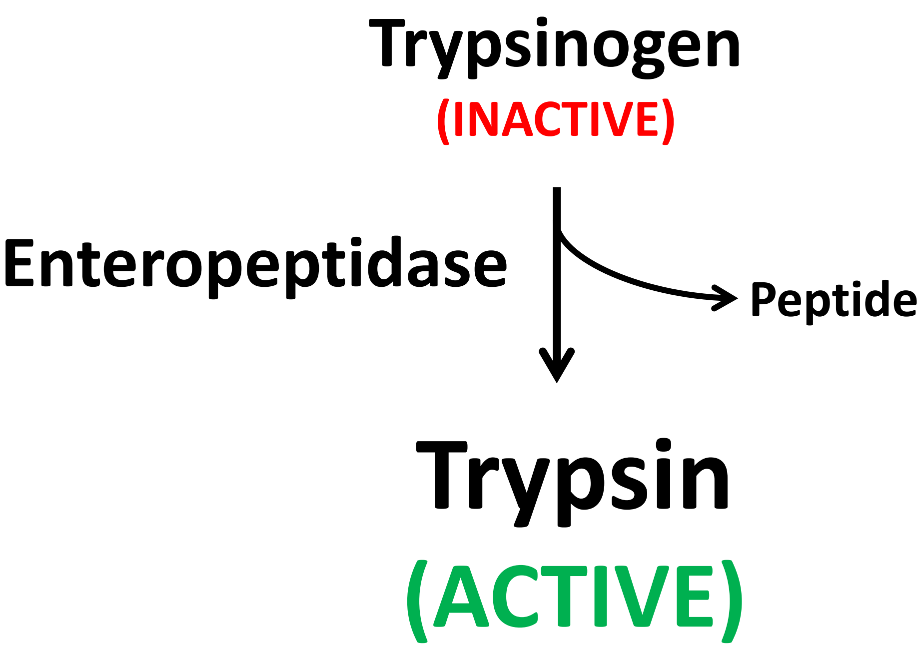


Feed-forward activation



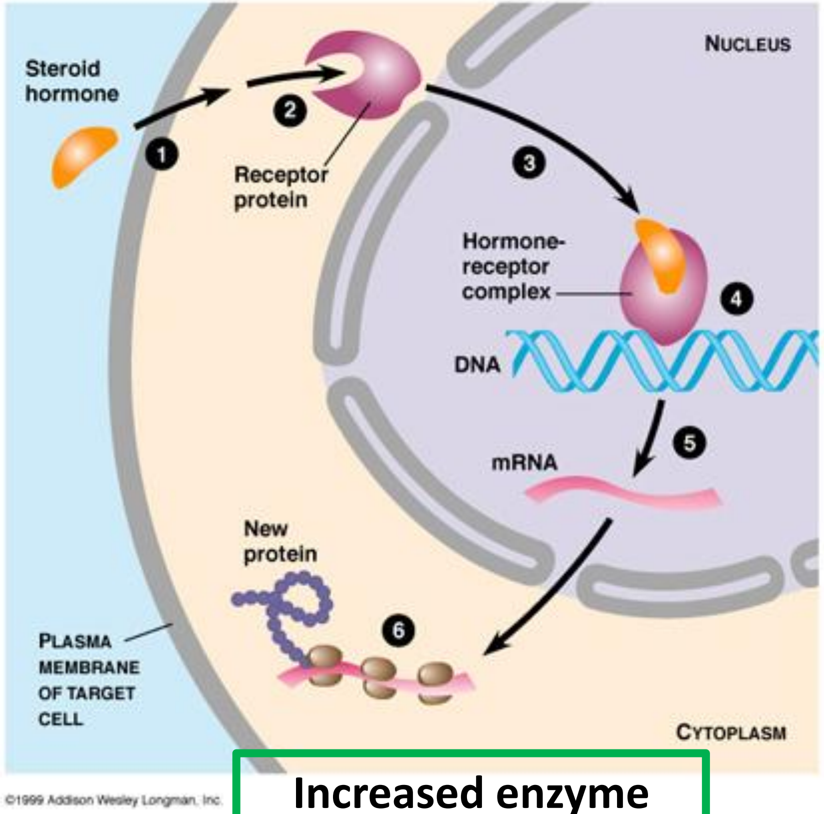
Short term regulation: **Irreversible proteolytic activation**

- Enzyme synthesized in inactive form
- Rapidly activated by specific stimuli when required
- A short peptide cleaved (Irreversible activation)
- Digestive enzymes
  - Pepsinogen → Pepsin by HCl (Hydrochloric acid) in stomach lumen
  - Trypsinogen (pancreas) → Trypsin in (duodenum) intestine by enteropeptidase
  - Chymotrypsinogen → Chymotrypsin in intestine by activated trypsin
- Clotting cascade proteins/enzymes activated in circulation



Long-term regulation: Induction (increased enzyme synthesis)

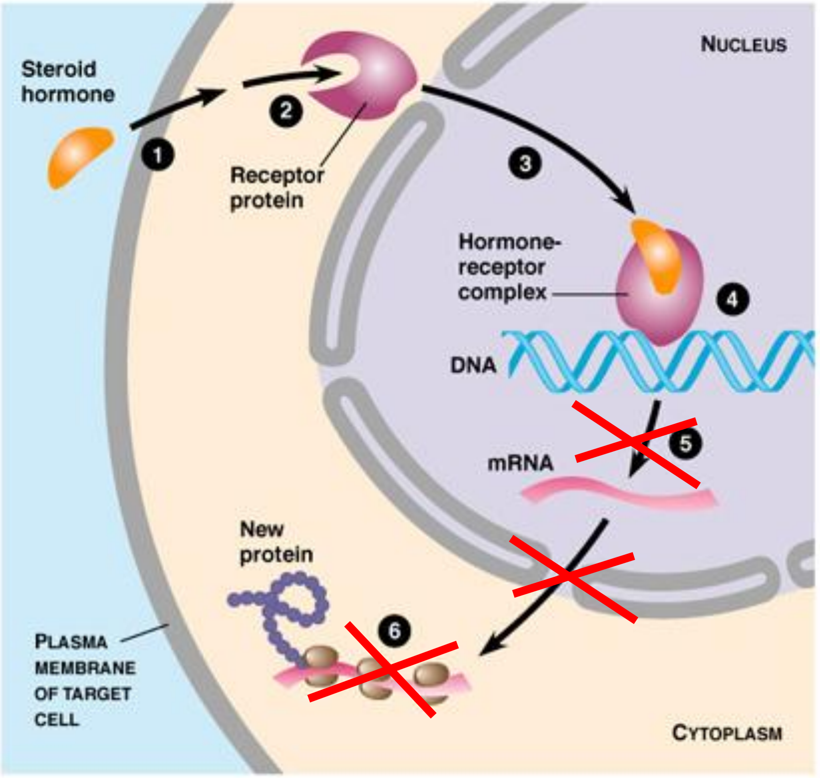
- Cells can alter rates of enzyme synthesis based on their requirement
- **Hours to days** to become effective (long-term)
- Examples: Steroid/ thyroid hormone family (refer signal transduction)
- **Inducer** (hormone/ other signals) → Hormone-receptor complex → Bind to DNA → Increase transcription of specific genes → Increased mRNA (Increased gene expression) → Increased Translation → Increased enzyme synthesis → Increased levels of enzyme **(Induction)**



Increased enzyme synthesis (Induction) → **Increased enzyme levels** → **Increased flux through the pathway**

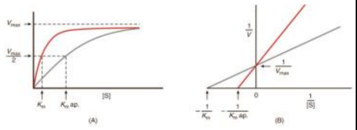
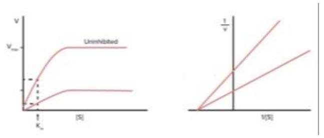
Long-term regulation: Repression (Decreased enzyme synthesis)

- Cells can alter rates of enzyme synthesis based on their requirement
- **Hours to days** to become effective (long-term)
- Examples: Steroid/ thyroid hormone family (refer signal transduction)
- **Repressor** (Hormone/ other signal) → Hormone-receptor complex (acts as repressor) → Binds to DNA → Decreases transcription of genes → Decreases mRNA formed (Decreased gene expression) → Decreased translation → Decreased synthesis of enzymes → Lower enzyme levels **(Repression)**

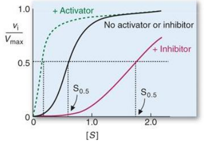


Decreased enzyme synthesis (Repression) → **Decreased enzyme levels** → **Decreased flux through the pathway**

Summary: Enzyme inhibition

Type of inhibition	Vmax	Km	Features of inhibition
Competitive (Reversible): Statin inhibits HMG CoA reductase	Unchanged 	Apparent Km increased	Inhibitor binds to active site and competes with substrate; Inhibition reversed at high [S]; Inhibitor is substrate analog; Inhibitor binds to enzyme reversibly
Non-competitive (Reversible) 	Apparent Vmax lower	Km unchanged	Inhibitor binds to site other than active site; Not reversed by high [S]; Inhibitor binds to enzyme reversibly
Irreversible inhibition (DFP inhibits acetylcholinesterase; Aspirin inhibits cyclooxygenase/ COX)	---	---	Inhibitor binds covalently to enzyme; Inhibition overcome by synthesis of NEW enzyme
Suicide inhibition (Allopurinol inhibits Xanthine oxidase)	--	--	Inhibitor binds to active site → Undergoes part of reaction and covalently (Irreversibly) binds to enzyme

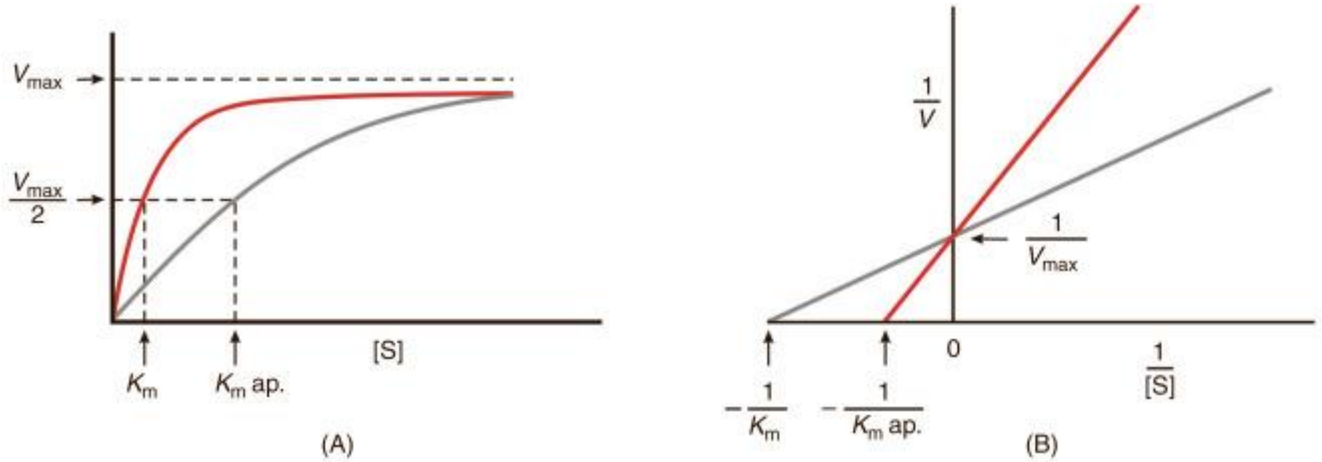
Summary: Enzyme regulation

Type of regulation	Kinetic changes	Features of regulation
Covalent modification	Short-term regulation	Phosphorylation/ dephosphorylation of enzyme → Conformational change of enzyme → Activation/ Less active enzyme
Allosteric regulation	Sigmoid (S-shaped curve) **K0.5 lower for allosteric activation** **K0.5 higher for allosteric inhibition** Short-term regulation	Modulator binds site other than active site → conformational change in enzyme **Allosteric activator → Increases Substrate binding** **Allosteric inhibitor → Decreases Substrate binding** 
Feed-forward (Allosteric) activation	Short-term regulation	**Substrate of pathway (Allosteric activator) → Increases Substrate binding → Increases flux through pathway**
Product inhibition (Feedback/ Allosteric inhibition)	Short-term regulation	**Product of pathway (Allosteric inhibitor) → Decreases Substrate binding → Decreases flux through pathway**
Proteolytic activation	Short-term regulation	Inactive enzyme converted to active enzyme by cleavage of a peptide
Induction	Long-term regulation	Stimulus (hormone) → **Increases gene expression** of the enzyme → Increased mRNA → Increased enzyme synthesis → Increased flux through pathway
Repression	Long-term regulation	Stimulus (hormone) → **Decreases gene expression** of the enzyme → Decreased mRNA → Decreased enzyme synthesis → Decreased flux through pathway

Self-study

- Case-study answers in practice questions on Sakai

Case study: 1



- Explain the type of inhibition and identify characteristics
- Identify changes in kinetics
- Draw a line to show changes when inhibitor concentration is increased
- Give examples: Drug and enzyme inhibited

Case study: 2

- Identify the types of inhibition in figures 1 and 2
- Identify the Michalis-Menten graphs for both types of inhibition
- Identify the kinetic changes in both types of inhibition
- Draw lines to show increasing concentration of the inhibitor in both the graphs

Figure 1

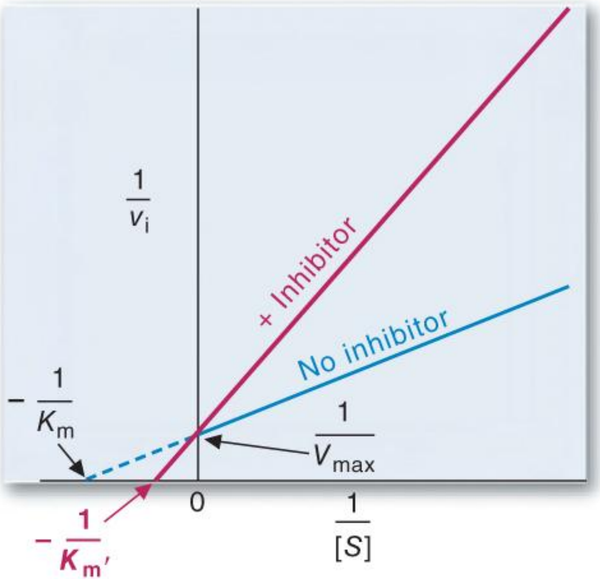
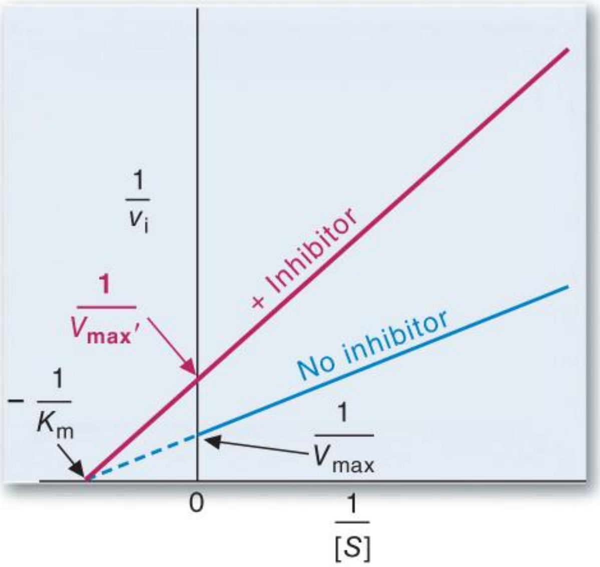


Figure 2



Case study 3

- Explain characteristics of the regulated enzyme: Cooperative Substrate binding; Short-term; conformational change
- Identify the effect of addition of activator
- Identify effect of addition of inhibitor
- Identify the graph on addition of product of the pathway
- Identify the graph on addition of substrate of the pathway
- Explain changes in enzyme kinetics on addition of activator and inhibitor